

Mohammed Abdullah's app requirements

Project Overview

The platform is a web-based solution that enables customers in Saudi Arabia to compare prices of electronic devices (e.g., laptops, TVs, smartphones) across multiple local and online electronic stores (e.g., Extra, Almanae, Jarir). Customers can search for a specific product, filter by brand, model, and specifications, and the platform will return results with prices from different stores. Customers can then be redirected to the store with the lowest or preferred price. Phase 1 focuses on growth and adoption, while Phase 2 introduces profitability through commissions from sales.

Functional Requirements

- **User Registration & Authentication:** Allow optional account creation via email, phone, or social login.
- **Guest Access:** Customers can search without creating an account.
- **Product Search:** Search by product category (e.g., TV, Laptop, Smartphone).
- **Brand & Model Filtering:** Narrow down results by brand, model, specifications.
- **Advanced Filtering:** By size, RAM, storage, resolution, color, etc.
- **Price Comparison Engine:** Compare product prices across integrated stores.
- **Sort & Filter Results:** Sort by lowest price, popularity, rating, or store name.
- **Store Integration:** APIs/web scraping to fetch real-time prices from stores like Extra, Jarir, Almanea.
- **Redirection to Store:** Click-through to redirect customer to the store website with selected product.
- **Product Detail Page:** Display specifications, images, and available stores with pricing.
- **Wishlist/Save Products:** Users can save products for later viewing.
- **Notification System:** Alert users when prices drop or when products are back in stock.
- **Multi-language Support:** Arabic and English interfaces.
- **Currency Handling:** Prices displayed in SAR.
- **Search Suggestions:** Auto-suggest products while typing.

- User Reviews & Ratings: Allow customers to rate stores and leave reviews.
- Comparison Table: Show side-by-side comparison of specifications and prices.
- Commission Tracking: Track successful redirections and purchases for future monetization.
- Analytics Dashboard (Admin): Monitor platform traffic, top searched items, and store performance.
- Store Onboarding Portal: Enable stores to register and provide product feeds directly.
- Favorites & Personalized Recommendations: Suggest products based on past searches and saved items.
- Mobile Responsiveness: Fully optimized for mobile and desktop browsers.
- Future Mobile App Support: Android/iOS applications.
- Barcode/QR Code Scanner: Allow scanning devices in-store to compare prices online instantly.
- Search by Voice: Voice-enabled search functionality in Arabic and English.
- Price History Tracking: Show historical price trends for products.
- Daily Deals Section: Highlight best deals and discounts from multiple stores.
- Email/SMS Alerts: Send notifications on new offers or sales.
- Comparison Across Stores: Highlight delivery charges, warranty differences, and return policies.
- Favorites Sync: Sync saved products across web and mobile app.
- Multiple Login Options: Google, Apple ID, Facebook, and email.
- Affiliate Tracking: Unique tracking links for each redirected sale.
- Product Availability Checker: Indicate if the product is in stock or out of stock at each store.
- Dynamic Search Filters: Based on the selected product category.
- Product Videos/Demos: Embedded YouTube or store-provided product videos.
- Store Rating System: Aggregate customer ratings of stores.
- Smart Recommendations: AI-based suggestions of similar products or better deals.
- Coupon Integration: Display available discount codes or offers for each store.

- Multi-Store Cart (Phase 2+): Ability to add products from multiple stores before redirecting.
- Comparison by Delivery Time: Show expected delivery durations alongside prices.
- Localized Store Integration: Tie-ins with Saudi-specific electronic chains and marketplaces.
- Gift Option Integration: Option to send products as gifts with links to stores.
- Search History Log: Keep track of previously searched items.
- Single Sign-On: OAuth support for corporate or academic accounts (optional).

Non-Functional Requirements

- Performance: Search results should load in under 3 seconds.
- Scalability: Ability to handle 1 million+ users and concurrent searches.
- Security: Protect user data with SSL, encryption, and secure login methods.
- Reliability: 99.9% uptime with automated monitoring.
- Data Accuracy: Ensure store data is refreshed regularly and reliably.
- Usability: Simple and intuitive UI for both Arabic and English users.
- SEO Optimization: Ensure discoverability on search engines.
- Accessibility: WCAG 2.1 compliance for users with disabilities.
- Maintainability: Modular codebase for easy updates and new store integrations.
- Legal Compliance: Align with Saudi e-commerce and data privacy laws.
- Cross-Browser Compatibility: Support Chrome, Safari, Edge, Firefox.
- Mobile Optimization: Smooth experience across iOS and Android devices.
- Data Backup: Daily automated backups of product data and user preferences.
- Monitoring & Logging: Full logging of API requests and error tracking.
- Energy Efficiency: Optimized infrastructure to reduce hosting costs.
- Localization: Support Hijri and Gregorian dates, Arabic numerals.

Phase 1 (Growth-Oriented)

- Build core search and comparison features.
- Integrate with at least 5 major stores (Extra, Jarir, Almanea, Noon, Amazon.sa).

- Support Arabic and English interfaces.
- Free service for all stores to encourage adoption.
- Collect traffic data and user behavior for optimization.
- Mobile responsiveness as a priority.
- Basic analytics dashboard for admin.
- SEO-friendly product catalog.

Phase 2 (Monetization-Oriented)

- Introduce commission model with stores for each sale.
 - Premium store listing options.
 - Personalized product recommendations.
 - Loyalty program or cashback system.
 - Mobile application launch with push notifications.
 - AI-based product recommendations.
 - Store dashboard for performance analytics.
 - Coupon and discount code integration.
 - Voice search and barcode scanner features.
- Scalability: The system MUST support more than five stores (no hard cap at 5). Design for horizontal scaling across data sources, tenants, and traffic spikes.
 - Phasing: Defer Voice search, QR code scanning, and the price history timeline to Phase 2 (exclude from MVP scope).
 - AI from Day 1: Include recommendation engine and customer behavior analytics in Phase 1. Baseline models must be productionized (even if simple) with data pipelines and monitoring.
 - Mobile App (Phase 1 build): Implement the mobile application during Phase 1 even if go-live is postponed. Keep feature parity with core web flows where feasible.
 - Integrations: Prefer official/partner APIs over web scraping for data ingestion. Scraping is a last resort with compliance, throttling backoffs, and maintenance budget.